

Challenge (Habanero)**Q1.** Solve the system: $x + y = 5$, $x^2 - y^2 = 7$. What is x ?

- (A) $x = 2$
 (B) $x = 3$
 (C) $x = 4$
 (D) $x = 5$
 (E) $x = 6$

Q2. In a game, a player's score S is given by $S = 2x^2 - 5x + 3$ and their level L is given by $L = x + 2$. If $S = 15$, what is L ?

- (A) $L = 2$
 (B) $L = 3$
 (C) $L = 4$
 (D) $L = 5$
 (E) $L = 6$

Q3. Solve the system: $y = 2x - 1$, $y = x^2 - 4x + 3$. What is y ?

- (A) $y = -2$
 (B) $y = -1$
 (C) $y = 0$
 (D) $y = 1$
 (E) $y = 2$

Q4. A gamer's data usage D is given by $D = x^2 + 2x + 1$ and their download speed S is given by $S = 3x - 2$. If $D = 16$, what is S ?

- (A) $S = 4$
 (B) $S = 5$
 (C) $S = 6$
 (D) $S = 7$
 (E) $S = 8$

Q5. Solve the system: $x^2 + y^2 = 25$, $x - y = 3$. What is x ?

- (A) $x = 2$
 (B) $x = 3$
 (C) $x = 4$
 (D) $x = 5$
 (E) $x = 6$

Q6. A programmer's code efficiency E is given by $E = 2x^2 - 3x + 1$ and their code speed S is given by $S = x + 1$. If $E = 5$, what is S ?

- (A) $S = 2$
 (B) $S = 3$
 (C) $S = 4$
 (D) $S = 5$
 (E) $S = 6$

Q7. Solve the system: $y = x^2 - 2x - 3$, $y = 2x - 1$. What is x ?

- (A) $x = -1$
 (B) $x = 0$
 (C) $x = 1$
 (D) $x = 2$
 (E) $x = 3$

Q8. A gaming company's revenue R is given by $R = x^2 + 2x + 1$ and their profit P is given by $P = 3x - 2$. If $R = 25$, what is P ?

- (A) $P = 5$
 (B) $P = 6$
 (C) $P = 7$
 (D) $P = 8$
 (E) $P = 9$

Open Ended Questions (FRQ)**Q9.** Solve the system: $x^2 + y^2 = 16$, $x - y = 2$. Find the intersection points and describe their significance in the context of a game's scoreboard.**Q10.** A programmer is designing a game where the player's score S is given by $S = x^2 - 2x + 1$ and their level L is given by $L = 2x - 1$. If $S = 20$, find L and interpret the result in the context of the game's logic.**Chef's Tips**

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- Tip 1: Emphasize the importance of checking solutions in the original system of equations.
- Tip 2: Encourage students to use graphing tools to visualize the intersection points.
- Tip 3: Remind students to consider the context of the problem when interpreting their results.